TSM 098 | INDEPENDENT PRACTICE

Prime factorization and numerical fractions

While TSM 099 studies Topic 18. Review: Topic 2, Topic 4.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: prime factorization

1. Write as a product of prime factors.

36

2. Write as a product of prime factors.

48

3. Write as a product of prime factors.

60

4. Write as a product of prime factors.

72

5. Write as a product of prime factors.

84

6. Write as a product of prime factors.

90

7. Write as a product of prime factors.

120

8. Write as a product of prime factors.

180

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Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: simplify using common prime factors

9. Write in lowest terms.

$$\frac{36}{84}$$

10. Write in lowest terms.

$$\frac{48}{180}$$

11. Write in lowest terms.

$$\frac{72}{108}$$

12. Write in lowest terms.

$$\frac{90}{150}$$

13. Write in lowest terms.

$$\frac{126}{210}$$

14. Write in lowest terms.

$$\frac{-84}{132}$$

15. Write in lowest terms.

$$\frac{144}{96}$$

16. Write in lowest terms.

$$\frac{-120}{-168}$$

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Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: equivalent fractions

17. Find the missing numerator.

$$\frac{3}{4} = \frac{?}{24}$$

18. Find the missing numerator.

$$\frac{5}{6} = \frac{?}{30}$$

19. Find the missing numerator.

$$\frac{-2}{9} = \frac{?}{36}$$

20. Find the missing numerator.

$$\frac{7}{10} = \frac{?}{30}$$

21. Find the missing numerator.

$$\frac{4}{15} = \frac{?}{60}$$

22. Find the missing numerator.

$$\frac{11}{12} = \frac{?}{36}$$

23. Find the missing numerator.

$$\frac{-5}{8} = \frac{?}{40}$$

24. Find the missing numerator.

$$\frac{0}{7} = \frac{?}{63}$$

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Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Extension: compare and explain

25. Are these fractions equivalent? Justify.

$\frac{6}{8}$ and $\frac{9}{12}$

26. Are these fractions equivalent? Justify.

$\frac{4}{9}$ and $\frac{8}{15}$

27. Are these fractions equivalent? Justify.

$\frac{-3}{5}$ and $\frac{9}{-15}$

28. Are these fractions equivalent? Justify.

$\frac{12}{18}$ and $\frac{3}{4}$

29. Write in lowest terms.

$\frac{252}{420}$

30. Write in lowest terms.

$\frac{-165}{225}$

31. Explain why dividing only the numerator by 3 changes the value.

$\frac{9}{12}$

32. Can multiplying both numerator and denominator by zero produce an equivalent fraction? Explain.

TSM 098 | ANSWER KEY

Prime factorization and numerical fractions

While TSM 099 studies Topic 18. Review: Topic 2, Topic 4.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: prime factorization

1. Write as a product of prime factors.

$36 = 2^2 \cdot 3^2$

2. Write as a product of prime factors.

$48 = 2^4 \cdot 3$

3. Write as a product of prime factors.

$60 = 2^2 \cdot 3 \cdot 5$

4. Write as a product of prime factors.

$72 = 2^3 \cdot 3^2$

5. Write as a product of prime factors.

$84 = 2^2 \cdot 3 \cdot 7$

6. Write as a product of prime factors.

$90 = 2 \cdot 3^2 \cdot 5$

7. Write as a product of prime factors.

$120 = 2^3 \cdot 3 \cdot 5$

8. Write as a product of prime factors.

$180 = 2^2 \cdot 3^2 \cdot 5$

TSM 098 | ANSWER KEY

Prime factorization and numerical fractions

While TSM 099 studies Topic 18. Review: Topic 2, Topic 4.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: simplify using common prime factors

9. Write in lowest terms.

$$\frac{36}{84} = \frac{3}{7}$$

10. Write in lowest terms.

$$\frac{48}{180} = \frac{4}{15}$$

11. Write in lowest terms.

$$\frac{72}{108} = \frac{2}{3}$$

12. Write in lowest terms.

$$\frac{90}{150} = \frac{3}{5}$$

13. Write in lowest terms.

$$\frac{126}{210} = \frac{3}{5}$$

14. Write in lowest terms.

$$\frac{-84}{132} = -\frac{7}{11}$$

15. Write in lowest terms.

$$\frac{144}{96} = \frac{3}{2}$$

16. Write in lowest terms.

$$\frac{-120}{-168} = \frac{5}{7}$$

TSM 098 | ANSWER KEY

Prime factorization and numerical fractions

While TSM 099 studies Topic 18. Review: Topic 2, Topic 4.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: equivalent fractions

17. Find the missing numerator.

$$\frac{?}{18}$$

Multiply numerator and denominator by 6.

18. Find the missing numerator.

$$\frac{?}{25}$$

Multiply numerator and denominator by 5.

19. Find the missing numerator.

$$\frac{?}{-8}$$

Multiply numerator and denominator by 4.

20. Find the missing numerator.

$$\frac{?}{21}$$

Multiply numerator and denominator by 3.

21. Find the missing numerator.

$$\frac{?}{16}$$

Multiply numerator and denominator by 4.

22. Find the missing numerator.

$$\frac{?}{33}$$

Multiply numerator and denominator by 3.

23. Find the missing numerator.

$$\frac{?}{-25}$$

Multiply numerator and denominator by 5.

24. Find the missing numerator.

$$\frac{?}{0}$$

Multiply numerator and denominator by 9.

TSM 098 | ANSWER KEY

Prime factorization and numerical fractions

While TSM 099 studies Topic 18. Review: Topic 2, Topic 4.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Extension: compare and explain

25. Are these fractions equivalent? Justify.

$\frac{3}{4}$ and $\frac{3}{4}$

Yes; the reduced values match.

26. Are these fractions equivalent? Justify.

$\frac{4}{9}$ and $\frac{8}{15}$

No; the reduced values differ.

27. Are these fractions equivalent? Justify.

$-\frac{3}{5}$ and $-\frac{3}{5}$

Yes; the reduced values match.

28. Are these fractions equivalent? Justify.

$\frac{2}{3}$ and $\frac{3}{4}$

No; the reduced values differ.

29. Write in lowest terms.

$\frac{252}{420} = \frac{3}{5}$

30. Write in lowest terms.

$\frac{-165}{225} = -\frac{11}{15}$

31. Explain why dividing only the numerator by 3 changes the value.

$\frac{9}{12} \neq \frac{3}{12}$

The original is 3/4; the new fraction is 1/4. Divide both parts by the same nonzero factor to preserve value.

32. Can multiplying both numerator and denominator by zero produce an equivalent fraction? Explain.

No. It produces 0/0, which is undefined.